

COURSE CODE- **CV374TGH**

USN-

RV COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

MODEL QUESTION PAPER

VII Semester BE February- March -2026 Examinations

DEPARTMENT OF CIVIL ENGINEERING

COURSE- Freight Transportation Systems and Logistics

Time-03 Hours

Maximum Marks-100

Instructions to Candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 or 4, 5 or 6 ,7 or 8 ,9 or 10.

PART A

Q No		Questions	Marks
1	1.1	Explain the term freight capacity issues with an example.	02
	1.2	How does freight productivity influence overall logistics performance?	02
	1.3	Briefly describe the role of scheduling in freight transportation.	02
	1.4	List any two current trends in logistics and explain them.	02
	1.5	What is the aim of logistics in a supply chain context?	02
	1.6	Define Freight Characteristics and list any three key attributes of freight transport.	02
	1.7	List the four main freight agents involved in freight demand estimation.	02
	1.8	State the primary purpose of freight transportation planning.	02
	1.9	Differentiate between Freight Generation (FG) and Freight Trip Generation (FTG) in one sentence each.	02
	1.10	Identify two major problems in urban freight transportation.	02

PART B

UNIT-I			
2	a	Analyze the factors influencing freight travel and explain with suitable examples how any three factors affect modal choice in intermodal freight transport.	08
	b	Analyze the key differences between passenger and freight demand models.	08

UNIT-II														
3	a	Explain the role of freight agents in the freight demand estimation process.	16											
	b	Discuss the characteristics of freight transport in regional and urban goods movement												
	OR													
4	a	A warehouse in an industrial zone handles 450 tons of goods daily. The freight trip rate is 0.75 trips per ton for outbound delivery. Calculate: a) Total freight trips generated per day. b) If each truck has an average payload of 12 tons, find the number of truck trips required.	08											
	b	A two-sector economy has the following data (in million Rupees): <table border="1"> <thead> <tr> <th></th><th>Agriculture</th><th>Manufacturing</th><th>Final Demand</th></tr> </thead> <tbody> <tr> <td>Agriculture</td><td>15</td><td>35</td><td>50</td></tr> <tr> <td>Manufacturing</td><td>20</td><td>60</td><td>120</td></tr> </tbody> </table> Assume value-to-weight ratio: • Agriculture: 1 ton = Rs. 1,000 • Manufacturing: 1 ton = Rs. 4,000 Estimate total freight generation (in tons) from final demand only.		Agriculture	Manufacturing	Final Demand	Agriculture	15	35	50	Manufacturing	20	60	120
	Agriculture	Manufacturing	Final Demand											
Agriculture	15	35	50											
Manufacturing	20	60	120											

UNIT-III			
5	a	Explain the factors affecting freight productivity and performance. Illustrate how organizations measure these factors.	08
	b	Describe the process of route planning, scheduling, collection, storage, and distribution center operations in logistics networks.	08
OR			
6	a	Evaluate the importance of distribution centers in improving freight performance. Provide examples of their strategic roles.	08
	b	Compare various competing modes of transport for specific commodities. How does mode selection affect cost and service levels?	08

UNIT-IV			
7	a	Explain strategic decisions in logistics and discuss the steps involved in designing an effective logistics strategy.	08
	b	Explain the concept of logistics strategy and describe the key strategic decisions involved in designing a logistics strategy.	08
OR			

8	a	Discuss the aims and importance of logistics in supporting organizational competitiveness.	08
	b	Summarize the current trends in logistics and explain how they influence modern supply chains.	08

UNIT-V			
9	a	Describe the importance of facility location in logistics and explain the factors considered while choosing a geographic region.	08
	b	Differentiate between Infinite Set Approaches and Feasible Set Approaches used in location planning.	08
OR			
10	a	Explain the types of planning in logistics and describe the role of capacity planning and capacity adjustment.	08
	b	Discuss the concept of sustainable and green logistics and explain how digital technologies such as IoT, big data analytics, AI, and blockchain support logistics management.	08

..... Q.P.Code

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PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	1	1	1	1	1	1	1	1	2	1
COs	2	1	4	3	2	1	1	3	3	2

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	3	1		3	a	3	2
	b	3	1			b	3	2
	c					c		
	d					d		
4	a	3	2		5	a	2	3
	b	3	2			b	3	3
	c					c		
	d					d		
6	a	3	3		7	a	2	3
	b	3	3			b	3	3
	c					c		
	d					d		
8	a	3	3		9	a	3	3
	b	3	3			b	2	3
	c					c		
	d					d		
10	a	3	3		11	a		
	b	3	3			b		
	c					c		
	d					d		

Signature of Scrutinizer:

Signature of Chairman

Name:

Name: